

James D. Motes

University of Illinois Urbana-Champaign
jnotes2@illinois.edu | jamesmotes.github.io

RESEARCH PROFILE & AGENDA

Robotics and AI researcher developing **planning systems that make large robot teams usable by non-experts**. My work shifts detailed robot programming from users to planning systems, accelerates those systems to practical engineering timescales, and exposes the task and coordination decisions people need to inspect and revise.

Scalable Robot-Team Planning: Multi-robot task and motion planning algorithms that coordinate robots only where conflicts, tasks, or workspace structure require it.

Accelerated Planning Infrastructure: Parallel, SIMD, and GPU-oriented planning methods that keep computation practical as robot teams and tasks grow in scale and complexity.

Human-Steerable Planning: Human-facing interfaces, such as natural language and AR/VR, that let people specify goals, understand team behavior and tradeoffs, and revise plans without programming individual robots.

ACADEMIC APPOINTMENT

Postdoctoral Researcher, University of Illinois Urbana-Champaign

August 2023 - Present

Advisor: Dr. Nancy M. Amato

EDUCATION

Ph.D. in Computer Science, University of Illinois Urbana-Champaign

August 2023

Ph.D. Thesis: Multi-robot Task and Motion Planning in Hybrid State Spaces

Advisor: Dr. Nancy M. Amato

M.S. in Computer Science, Texas A&M University

August 2019

M.S. Thesis: Interaction Templates for Multi-Robot Systems

Advisor: Dr. Nancy M. Amato

B.S. in Computer Engineering, Texas A&M University

May 2018

Minor in Mathematics, Engineering Honors, Undergraduate Research Scholar

RESEARCH LEADERSHIP HIGHLIGHTS

- Lead several graduate research teams in the Parasol Lab across multi-robot planning, accelerated planning, human-steerable planning, and open-source planning infrastructure.
- Set project directions, coordinate research milestones, mentor paper development, and integrate related projects into a coherent agenda around scalable planning for robot teams.
- Mentor over 20 graduate students, including six now graduated and five now PhD candidates; mentoring has contributed to 10 publications and another 3 now in review.
- Primary author on NSF and industry collaboration research proposals.
- Coordinate collaborations spanning robotics, NLP, HRI, computational geometry, industrial automation, and planning systems.

SELECTED TECHNICAL CONTRIBUTIONS

Decomposable Multi-Robot Planning: Developed and extended hypergraph-based planning frameworks for multi-robot task and motion planning, including DaSH[P5], Lazy-DaSH[P14], and guidance informed hypergraph planning[P12].

Adaptive Robot Coordination: Developed multi-robot motion planning strategies that dynamically couple and decouple robots around local coordination conflicts[P7], with extensions to kinodynamic planning[P10], experience reuse[U5], parallelization[P15], and asymptotic optimality[U2].

Accelerated Planning: Developed methods for accelerating planning using computational geometry[P8], lazy validation[P14], multicore CPU parallelization[P15], and GPU/SIMD serialization[U1,U4] for dynamic and multi-robot environments.

Human-Steerable Planning Interfaces: Developed planning tools that expose planning controls to non-expert users through interfaces like natural language[IP2] and AR/VR systems[U3].

PUBLICATIONS & SCHOLARLY OUTPUTS

Publication Overview

- 15 peer reviewed archival publications, including 12 journal articles and 3 refereed conference proceedings.
- 4 first author peer reviewed publications and 2 first author manuscripts under review.
- 10 student-led publications directly mentored with another 3 under review and 4 in preparation.
- Additional outputs include 11 preprints, technical reports, workshop papers, and extended abstracts.
- Google Scholar: 397 citations, h-index: 7, as of 08/10/2026.

Notation

- Name is bolded: **James D. Motes**
- * indicates equal contribution
- † indicates a student or junior researcher directly mentored by Motes: A. Student[†]
- ‡ indicates a student-led paper for which Motes served as primary day-to-day research mentor or project lead: **James D. Motes**[‡]

Research Theme Map

- Scalable Robot-Team Planning: [P1-5], [P7], [P10], [P12-14], [U2], [U5], [IP3-4]
- Accelerated Planning Infrastructure: [P4], [P8], [P15], [U1], [U4]
- Human-Steerable Planning: [U3], [IP1-2]

Peer-Reviewed Archival Publications

Journal Articles

- [P15] **James D. Motes**, M. Morales, N. M. Amato, "P-ARC: Exploiting Subproblem Independence for Parallel Multi-Robot Motion Planning," *IEEE Robotics and Automation Letters (RA-L)*, conditionally accepted, 2026.
- [P14] S. Lee[†], **James D. Motes**[‡], I. Ngui[†], M. Morales, and N. M. Amato, "Lazy-DaSH: Lazy Approach for Hypergraph-based Multi-Robot Task and Motion Planning," *IEEE Transactions on Robotics (T-RO)*, to appear, 2026.
- [P12] C. McBeth[†], **James D. Motes**[‡], I. Ngui[†], M. Morales, and N. M. Amato, "Scalable Multi-Robot Motion Planning Using Guidance-Informed Hypergraphs," *IEEE Robotics and Automation Letters (RA-L)*, 2026.
- [P11] H. Lee[†], **James D. Motes**[‡], M. Morales, and N. M. Amato, "An Analysis of Constraint-Based Multi-Agent Pathfinding Algorithms," *IEEE Transactions on Robotics (T-RO)*, 2025. Presented at the *IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, June 2026.
- [P10] M. Qin[†], I. Solis[†], **James D. Motes**[‡], M. Morales, and N. M. Amato, "K-ARC: Adaptive Robot Coordination for Multi-Robot Kinodynamic Planning," *IEEE Robotics and Automation Letters (RA-L)*, 2025. Presented at the *IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, June 2026.
- [P7] I. Solis[†], **James D. Motes**[‡], M. Qin[†], M. Morales, and N. M. Amato, "Adaptive Robot Coordination: A Subproblem-based Approach for Hybrid Multi-Robot Motion Planning," *IEEE Robotics and Automation Letters (RA-L)*, 2024. Presented at *IEEE ICRA@40*, Rotterdam, The Netherlands, Sept. 2024.
- [P6] C. McBeth[†], **James D. Motes**[‡], D. Uwacu, M. Morales, and N. M. Amato, "Scalable Multi-Robot Motion Planning for Congested Environments With Topological Guidance," *IEEE Robotics and Automation Letters (RA-L)*, 2023. Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, May, 2024.
- [P5] **James D. Motes**, T. Chen, T. Bretl, M. M. Aguirre, and N. M. Amato, "Hypergraph-Based Multi-Robot Task and Motion Planning," *IEEE Transactions on Robotics (T-RO)*, 2023. Presented at the *IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, May 2024.
- [P4] H. Lee[†], **James D. Motes**[‡], M. Morales, and N. M. Amato, "Parallel Hierarchical Composition Conflict-Based Search for Optimal Multi-Agent Pathfinding," *IEEE Robotics and Automation Letters (RA-L)*, vol. 6, no. 4, pp. 7001–7008, 2021. Presented at *IEEE International Conference on Intelligent Robots and Systems (IROS)*, Prague, Czech Republic, 2021 (virtual).
- [P3] I. Solis[†], **James D. Motes**[‡], R. Sandström, and N. M. Amato, "Roadmap-Optimal Multi-Robot Motion Planning Using Conflict-Based Search," *IEEE Robotics and Automation Letters (RA-L)*, vol. 6, no. 3, pp. 4608–4615, 2021. Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Xi'an, China, 2021 (virtual).
- [P2] **James D. Motes**, R. Sandström, H. Lee[†], S. Thomas, and N. M. Amato, "Multi-Robot Task and Motion Planning With Subtask Dependencies," *IEEE Robotics and Automation Letters (RA-L)*, vol. 5, no. 2, pp. 3338–3345, 2020. Presented at *IEEE International Conference on Robotics and Automation, Paris (ICRA)*, France, 2020 (virtual).
- [P1] **James D. Motes**, R. Sandström, W. Adams, T. Ogunyale[†], S. Thomas, and N. M. Amato, "Interaction Templates for

Multi-Robot Systems,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 4, no. 3, pp. 2926–2933, 2019.
Presented at *IEEE International Conference on Intelligent Robots and Systems (IROS)*, Macao, China, 2019.

Refereed Conference Proceedings

- [P13] I. Ngui[†], C. McBeth[†], **James D. Motes[‡]**, M. Morales, and N. M. Amato, “Scalable Multi-Robot Motion Planning via Hierarchical Subproblem Expansion and Workspace Decomposition Refinement,” in *Algorithmic Foundations of Robotics XVII: Proc. the Seventeenth World Symposium on the Algorithmic Foundations of Robotics (WAFR)*, 2026.
- [P9] A. Goddu, T. Brown, M. Paton, **James D. Motes**, and T. Chen, “Modifying ABIT* for Tethered Rappelling Robot Motion Planning,” in *Proc. 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, pp. 722–728, 2025.
- [P8] S. Ashur[†], M. Lusardi, M. Markowicz[†], **James D. Motes[‡]**, M. Morales, S. Har-Peled, and N. M. Amato, “SPITE: Simple Polyhedral Intersection Techniques for Modified Environments,” in *Algorithmic Foundations of Robotics XVI: Proc. the Sixteenth Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2024.

Manuscripts Under Review

First Author

- [U2] **James D. Motes**, M. Morales, N. M. Amato, “AO-ARC: Almost-Surely Asymptotically Optimal Multi-Robot Motion Planning with ARC.”
- [U1] **James D. Motes**, M. Morales, N. M. Amato, “Multi-Robot Motions in Milliseconds: Vector-Accelerated Primitives for Sampling-Based Planning,” *arXiv preprint arXiv:2604.23960*, 2026.

Mentored Student-Led Manuscripts

- [U5] I. Solis[†], **James D. Motes[‡]**, J. Montalvo, M. Qin[†], M. Morales, and N. M. Amato, “Experience-Based Subproblem Planning for Multi-Robot Motion Planning,” *arXiv preprint arXiv:2411.08851*, 2024.
- [U4] Y. Arad[†], M. Markowicz[†], S. Ashur[†], **James D. Motes[‡]**, M. Morales, and N. M. Amato, “Serialized Red-Green-Gray: A Quick Heuristic Validation of Edges in Dynamic Roadmap Graphs.”
- [U3] I. Ngui[†], C. McBeth[†], A. Santos, G. He, K. J. Mimnaugh, **James D. Motes**, L. Soares, M. Morales, and N. M. Amato, “ERUPT: An Open Toolkit for Interfacing with Robot Motion Planners in Extended Reality,” *arXiv preprint arXiv:2510.02464*, 2025.

Manuscripts in Preparation

- [IP4] S. Lee[†], **James D. Motes[‡]**, M. Morales, and N. M. Amato, “PDDL-DaSH: Expanding Hypergraph Planning With Explicit and Implicit Query Strategies.”
- [IP3] M. Markowicz[†], **James D. Motes[‡]**, I. Solis[†], M. Morales, and N. M. Amato, “MR-SPITE: Leveraging Dynamic Roadmap Graphs in Multi-Robot Motion Planning.”
- [IP2] A. Yammanuru[†], J. Buabeng[†], J. Liu[†], A. Luo[†], **James D. Motes[‡]**, Marco Morales, and Nancy M. Amato, “Object-centric Reasoning for Behavior with Instruction-conditioned Trajectories.”
- [IP1] I. B. Love[†], D. J. Lohan, **James D. Motes**, E. M. Dede, M. Morales, and N. M. Amato, “A Hierarchical Approach for the Generation of Varied Cable Harness Designs.”

Additional Scholarly Outputs

Preprints and Technical Reports

- [TR2] H. Lee[†], Z. Serlin, **James D. Motes**, B. Long, M. Morales, and N. M. Amato, “PRISM: Complete Online Decentralized Multi-Agent Pathfinding With Rapid Information Sharing Using Motion Constraints,” *arXiv preprint arXiv:2505.08025*, 2025.
- [TR1] A. Attali, S. Ashur[†], I. B. Love[†], C. McBeth[†], **James D. Motes**, D. Uwacu, M. Morales, and N. M. Amato, “A Framework for Guided Motion Planning,” *arXiv preprint arXiv:2404.03133*, 2024.

Workshop Papers and Extended Abstracts

- [W9] M. Markowicz[†], S. Ashur[†], **James D. Motes[‡]**, and N. M. Amato, “Semi-Lazy Rearrangement Solver for Accelerating Task Planning Using SPITE Dynamic Roadmaps,” in *Extended Abstract, 4th Workshop on Future of Construction: Safe, Reliable, and Precise Robots in Construction Environments (ICRA 2025)*, Atlanta, GA, USA, May 19, 2025.
- [W8] H. Lee[†], **James D. Motes[‡]**, Z. Serlin, M. Morales, and N. M. Amato, “Distributed Constraint-Based Search Using Multi-Hop Communication,” in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- [W7] I. Ngui[†], **James D. Motes[‡]**, H. Lee[†], M. Morales, and N. M. Amato, “Ergonomic Motion Planning for Human-Robot Collaboration,” in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- [W6] K. Elimelech,^{*} **James D. Motes[‡]**,^{*} M. Morales, N. M. Amato, M. Vardi, and L. E. Kavraki, “Encoding Reusable Multi-Robot Planning Strategies as Abstract Hypergraphs,” in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- [W5] S. Lee[†], **James D. Motes[‡]**, I. Ngui[†], M. Morales, and N. M. Amato, “Lazy-DaSH: Lazy Approach for Hypergraph-based Multi-Robot Task and Motion Planning,” in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.

- [W4] M. Lusardi,* M. Markowicz[†],* S. Ashur[†], **James D. Motes**, M. Morales, S. Har-Peled, and N. M. Amato, "SPITE: Simple Polyhedral Intersection Techniques for Modified Environments," in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- [W3] I. Ngui[†], S. Lee[†], **James D. Motes**[‡], M. Morales, and N. M. Amato, "A Hierarchical Approach to Workstation-based Task-Allocation and Motion Planning," in *Extended Abstract, RAFF2023: Robotics & AI in Future Factory (IROS 2023 Workshop)*, IROS 2023, Detroit, MI, USA, Oct. 1–5, 2023.
- [W2] A. Attali, S. Ashur[†], I. B. Love[†], C. McBeth[†], **James D. Motes**, D. Uwacu, M. Morales, and N. M. Amato, "Evaluating Guiding Spaces for Motion Planning," in *Extended Abstract, Evaluating Motion Planning Performance, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2022)*, Kyoto, Japan, Oct. 23, 2022.
- [W1] T. Chen, Z. Huang, **James D. Motes**, J. Geng, Q. Ta, H. Dinkel, H. Abdul-Rashid, J. Myers, Y. Mun, W. Lin, Y. Huang, S. Liu, M. Morales, N. M. Amato, K. Driggs-Campbell, and T. Bretl, "Insights from an Industrial Collaborative Assembly Project: Lessons in Research and Collaboration," in *IEEE ICRA Workshop on Collaborative Robots and the Work of the Future*, Philadelphia, USA, 2022.

PROPOSAL & SPONSORED RESEARCH DEVELOPMENT

National Science Foundation (NSF) - Foundational Research in Robotics (FRR) Proposal September 2025

- Primary author of proposal to customize robotic motion behaviors for non-expert users through natural language preference specification (Status: Pending)
- Coordinated collaboration with natural language processing researchers at the University of Illinois Urbana-Champaign
- Developed research narrative connecting human preference specification, motion planning, and human-steerable robot behavior.

National Science Foundation (NSF) - Engineering Research Center (ERC) Site Visit March 2026

- Supported preparation for an NSF Engineering Research Center site visit by developing research-thrust narrative and presentation materials.

Industry Proposals and White Papers April 2025 - Present

- Primary author of eight proposals and secondary author on four more
- Proposal topics include human-robot interaction, multi-robot planning, environment design, and data-driven planning
- Developed sponsored research narratives connecting scalable robot-team planning to deployable industrial automation systems

INVITED TALKS & SEMINARS

Lessons from Manufacturing Startups

- AI Meets Robots & Machines Seminar, University of Illinois Urbana-Champaign, Urbana, IL.
- February 2026. Invited seminar.

Multi-Robot Task and Motion Planning in Hybrid State Spaces

- Kavraki Lab, Department of Computer Science, Rice University, Houston, TX.
- February 2024. Invited Lab Seminar.

Planning Motions and Tasks for Manipulators, Multi-Robot Systems and Biomolecules

- Center for Autonomy Portfolio Discovery Workshop, University of Illinois Urbana-Champaign, Urbana, IL.
- October 2019. Invited Workshop Talk.

TEACHING & CURRICULUM DEVELOPMENT

Lead Instructor, AI4ALL August 2021 - December 2023

- Instructor for introductory artificial intelligence courses delivered through partner programs at the University of Texas at El Paso, New Mexico State University, Worcester Polytechnic Institute, and Texas A&M University.
- Developed curriculum for the Discover AI introductory artificial intelligence course.
- Led and mentored instructor teams across participating universities.
- Teaching assistant for Discover AI at the University of Illinois Urbana-Champaign in Fall 2021.

Teaching Assistant, University of Illinois Urbana-Champaign

January 2021 - May 2021

- Teaching assistant for Computer Science Ph.D. seminar course.
- Organized guest speakers, managed grading, and coordinated student assignments.

Internal Lab Instruction, Parasol Lab

August 2019 - May 2021

- Developed and taught a motion-planning course for new Parasol Lab members.

MENTORING, RESEARCH SUPERVISION & OUTREACH

Postdoctoral Researcher, Parasol Lab, University of Illinois Urbana-Champaign

August 2023 - Present

- Mentor over 20 graduate students across multi-robot planning, guided planning, accelerated planning, and human-steerable planning projects.
- Supported 6 graduate student completions and 5 current Ph.D. candidates.
- Postdoctoral mentoring has contributed to 6 student-led publications and another 3 manuscripts under review.
- Coordinate student research milestones, paper development, software integration, and cross-project planning infrastructure.

Graduate Researcher, University of Illinois Urbana-Champaign, Texas A&M University

August 2018 - August 2023

- Mentored 14 graduate students within the Parasol Lab research group (several continued under my mentorship during my postdoctoral appointment).
- Mentored undergraduate researchers, including local university students and DREU program participants.
- Graduate student mentoring contributed to 4 student-led publications.
- Taught motion planning to new lab members through an internal Parasol Lab course.

Houston Robotics Club

August 2023 - Present

- Mentor community college and high school students on AI and robotics projects.
- Supported students who later enrolled in STEM undergraduate programs or obtained internships with local robotics companies.

PROFESSIONAL SERVICE

Program Committee Member

- World Symposium on Algorithmic Foundations of Robotics, 2026.
- World Symposium on Algorithmic Foundations of Robotics, 2024.

Reviewer/Referee

August 2018 - Present

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Robotics (T-RO)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Symposium on Multi-robot and Multi-agent Systems (MRS)
- World Symposium (formerly International Workshop) on Algorithmic Foundations of Robotics (WAFR)
- International Journal of Robotics Research (IJRR)
- Artificial Intelligence Journal (AIJ)

Professional Memberships

- IEEE Robotics and Automation Society

May 2019 - Present

INDUSTRY, ENTREPRENEURSHIP & RESEARCH TRANSLATION

Industrial Human-Robot Factory Collaboration, UIUC

August 2019 - August 2023

- Collaborated with Foxconn Interconnect Technologies and UIUC research groups led by Timothy Bretl and Katherine Driggs-Campbell on collaborative human-robot factory projects.
- Connected multi-robot task and motion planning research to industrial assembly and human-robot collaboration settings.

Normandy Automation, Founder

May 2023 - August 2025

- Developed autonomous robotic manufacturing concepts for just-in-time manufacturing with minimal human reprogramming.
- Leveraged AI and multi-robot systems for autonomous welding and CNC machine-tending workflows.

- Wrote NSF SBIR proposals for autonomous robotic manufacturing systems.
- Optigon Inc., Principal Software Engineer, part time** May 2024 – Present
- Lead software development for DOE SBIR-funded high-throughput, non-contact spectroscopic metrology and analysis tools for clean-energy materials, including perovskite photovoltaics.
 - Develop machine learning analytical models of solar cell behavior.

HONORS

Engineering Graduate Merit Fellowship	August 2018 - August 2019
Presidential Endowed Scholarship	August 2014 - May 2018
Distinguished Student Award - Dwight Look College of Engineering	May 2017
Industrial Affiliates Program Scholarship	August 2014 - May 2018
Dell Merit Scholarship	August 2017 - May 2018